

DSL V - Z P D I

Mission Statement

Revision 3.6 · 28 September 2026 · Penrose, Colorado

We are building a way to listen for residual pairwise carrier-phase coherence after everything standard physics already predicts has been subtracted. Most of physics treats empty space as a quiet stage. That account works in scattering and precision spectroscopy. It is not the claim of this program. The claim is an upper bound.

The instrument is a distributed array of GPS-disciplined software-defined radio receivers, locked to a common GNSS time scale, tracking the same GPS L1 and L5 broadcast carriers from real ground. If an anomalous phase coupling survived Helmholtz propagation, multipath, clocks, common-view GNSS products, troposphere, and ionospheric gradients, it would appear as a faint inter-site correlation that the pre-registered null does not contain. Not a flash. Not a bang. A number on residual coherence.

We do not expect to find it. That is the point. This is a null experiment in the tradition of Michelson-Morley and the Fermilab Holometer. We measure the silence and put a hard number on it. If nothing survives the kill switches, we publish that number and the universe stays quiet — and that is a real result, not a failure. If something does survive every test, we do not jump to an ontology. We ask what it is, how strong it is, which chain it lives on, and whether any ordinary explanation was missed.

Two hypotheses, two bounds, neither standing in for the other. Chain S tests a sky-gradient residual and is blind to isotropic site-common phase. Chain C tests correlated site-common phase with a non-vanishing inter-site differential; a phase literally identical at every antenna is unobservable by construction. A null on one chain does not constrain the other.

The mission is to be honest enough to trust either answer, and to report three distinct columns every time: predicted budget | measured residual floor | reported bound.

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Document packet: mission cover · instrument white paper · field-book install and user guide (Appendix A).
Interpretive supplements S1/S2 are withheld from this submission.

Upper Bounds on Anomalous Distributed Carrier-Phase Coherence from a GNSS Receiver Array

A pre-registered instrument program with kill switches

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Document class: preprint prepared for peer-review submission

Date: 28 September 2026 · Revision 3.6 (submission compilation of Rev 3.5)

Changes implemented for this submission (Rev 3.5 → 3.6)

- Title and claims remain bounds. Vacuum structure is the program's motivating question, not a result. §§1–8 make no vacuum-ontology claim.
- All working equations are restored as selectable text (the Rev 3.5 Chrome/Skia PDF had rasterized math with an empty text layer).
- Supplements S1 (interpretive ontology) and S2 (companion biological/QED notes) are withheld. They are not part of the review claims and are not in this packet.
- Former kill switches 4 (Möbius holonomy) and 5 (gravitating plenum) are withdrawn with S1. Instrument switches are 1, 2, 3, and 6 only.
- $\Gamma_{\text{ext}}(t)$ is removed from the instrument symbol table. Optional environmental covariates may be logged in the field book; they are not detection inputs.
- Mission statement aligned to the bounds language so the cover and the paper do not disagree.
- Field book included as Appendix A (campaign record / install guide), not as a second science paper.
- Reference list restricted to sources used by §§1–8. S1/S2-only citations removed.
- Phase-0 validation is stated as a prerequisite, not a result. No measured floor is implied by the catalog budget.

Abstract

We describe a distributed array of GPS-disciplined software-defined-radio receivers (DSLVS-ZPDI) that sets upper bounds on anomalous pairwise carrier-phase coherence on the GPS L1/L5 broadcast carriers after subtraction of electromagnetic propagation (Helmholtz), site multipath, clock terms, common-view GNSS errors, tropospheric delay, and ionospheric gradients. Two processing chains are distinct by construction. Chain S uses between-node double differences of between-satellite single differences; satellite and receiver clocks cancel exactly, including the ≈ 0.7 rad shared IGS clock residual at L1 that would otherwise dominate inter-node coherence. It tests direction-differential (sky-gradient) residuals (H_S) and is blind to isotropic site-common phase. Chain C uses same-satellite inter-node differences; it keeps receiver-common phase and tests correlated site-common phase with non-vanishing inter-site differential (H_C) — a literally identical global phase being unobservable by construction — with receiver clocks carried by IGS products, measured GPSDO stability, and a pre-registered common-clock calibration whose transferability is explicitly bounded.

The per-node tracking product is NCO carrier phase plus prompt-discriminator residual at dump rate f_d , with data-bit wipeoff and pre-registered cycle-slip handling. Both analyses

consume two named input series on disjoint node pairs (≥ 4 nodes for detection-grade work). Analysis A bounds stationary residual coherence against a closed-form Beta null and a moving-block bootstrap; its floor is set by the effective number of independent samples, not calendar seconds. Analysis B forms a lag-resolved cross-correlation with bootstrap-max multiplicity control and lag resolution f_d^{-1} ; the $|\tau| \ll \tau_c$ morphology is claimed only on pair-of-pairs baselines with $\tau_c \gg f_d^{-1}$. Analysis A runs continuously and selects windows for deep Analysis-B follow-up; the joint false-alarm rate is calibrated end-to-end by time-slides on post-subtraction residuals and by blind injections into realistic residuals. A worked dual-frequency L-band budget shows the run is systematics-limited, with a per-snapshot double-difference phase floor of $\approx 0.4\text{--}0.8$ rad set by troposphere and multipath. This program specifies no coupling model and predicts no amplitude: it is a null experiment. Four instrument kill switches are stated, including a chromatic (uncombined L1/L5) falsification pathway that is necessary but not sufficient for any candidate excess. A null is the expected, publishable result: two upper bounds, one per chain, reported as predicted budget | measured floor | reported bound.

Keywords: distributed metrology, carrier-phase coherence, GPS-disciplined oscillators, software-defined radio, magnitude-squared coherence, null experiments, pre-registration, upper bounds

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Data availability, competing interests, and references

Appendix A. DSLV-ZPDI Field Book — Install and User Guide

Instrument program: DSLV-ZPDI — the program's distributed GPS-disciplined SDR receiver array. All "array" references herein denote DSLV-ZPDI nodes unless stated otherwise.

Epistemic legend. [E] Established experimental or textbook result. [I] Interpretive synthesis; not required by the data. [H] Hypothesis of this program. Contingent. Kill-switched.

1. The operational question

Textbook QED on Minkowski space treats the vacuum as the Poincaré-invariant ground state and localized excitations as particles. That account works in scattering and precision spectroscopy [E]. This program does not replace it, and this paper makes no vacuum-ontology claim. The instrument program DSLV-ZPDI is motivated by the vacuum-structure question; the paper's claims are upper bounds.

Operational question. After subtracting everything standard physics predicts — propagation, multipath, clocks, common-view GNSS, troposphere, ionospheric gradients, and catalogued transmitters — does a residual pairwise phase coherence remain that is inconsistent with the pre-registered null?

That question is not unique. An isotropic phase common to every tracked carrier at one antenna is cancelled by a between-satellite single difference. A satellite-common term is cancelled by a same-satellite inter-node difference. The program therefore states two observables and two bounds (§3).

Two hypotheses, named so the bounds mean something.

H_S: an anomalous phase coupling that depends on propagation direction — a sky gradient. Tested by Chain S, which cancels isotropic site-common phase by construction.

H_C: an anomalous phase component common to all tracked carriers at each site and correlated across sites, with non-vanishing inter-site differential. A phase literally identical at every antenna drops out of every inter-node difference and is unobservable by construction; H_C is the correlated differential that survives. Tested only by Chain C, which keeps receiver-common phase and therefore inherits the receiver-clock problem (§§3.1, 4.4, 5).

The two chains are complementary and neither subsumes the other: a null on Chain S does not constrain H_C, and a null on Chain C does not constrain H_S. The title hypothesis lives on the harder chain. That asymmetry is the point of naming them.

This program specifies no coupling model and predicts no amplitude. It is a null experiment in the tradition of Michelson-Morley and the Holometer [E]: it bounds anomalous coherence, and a null is the expected, publishable result. If an excess survived every switch, the next step would not be an ontology paper but a quantitative signal model — amplitude, frequency dependence, baseline dependence, chain response — demonstrating that no conventional mechanism reproduces it. Data before ontology.

Carrier-phase GNSS already treats phase as the primary observable; RTK resolves cycles to millimetre precision [E]. This program extends that practice from positioning to a null search on distributed residuals. The array is a phase microscope pointed at zero. What counts as interesting is defined in §§4-6. Everything else is scaffolding.

2. Prior art and the gap

Holometer (Fermilab). Closest interferometric prior art: two co-located 39 m Michelson interferometers, cross-spectrum to tens of MHz, including frequencies above the inverse

light-crossing time of the apparatus. First published sensitivity 2.1×10^{-20} m/ $\sqrt{\text{Hz}}$ after 2×10^8 independent spectra; nulls reported on holographic shear/length noise at that floor [E] (Chou et al., *Phys. Rev. Lett.* 117, 111102 (2016); instrument paper arXiv:1611.08265).

Difference, stated so a reviewer does not have to reconstruct it: the Holometer is co-located optical strain at one site. This program is RF carrier-phase residuals across geographic baselines at $\sim 10^3$ MHz. A null here constrains a different observable class. It does not repeat or refute the Holometer.

Clock networks and GNSS as exotic sensors. A second line of prior art treats GNSS constellations and terrestrial clock networks as sensors for physics beyond the Standard Model. Derevianko & Pospelov proposed the GPS atomic-clock constellation as a detector of topological dark matter [E], and Roberts et al. placed limits on domain-wall couplings using GPS satellite clock comparisons [E]. Those searches look for transient, propagating frequency modulation in clock comparisons. The present program's observable class — stationary, spatially distributed carrier-phase coherence after geodetic subtraction — is complementary: it requires no transient and compares phases, not frequencies. The two literatures do not constrain each other's channels, and a bound here is new information.

Methodological prior art. Morphology-agnostic detection with proper trials factors is standard practice in gravitational-wave burst searches. This program adopts the excess-power family of statistics (Anderson et al. 2001) and time-slide background estimation (e.g., Abadie et al. 2012) for Analysis B and for the joint Analysis-A $\rightarrow$ B false-alarm calibration [E].

Coherence statistic, not a dynamical model. Pairwise phase coherence on a window ΔT is used as a detection statistic only. Kuramoto/Strogatz are cited for the statistic, not as coupled-oscillator dynamics of the hardware [E]. See §3.4 for the one sense in which the nodes are not independent — the GPSDO steering band — and how that band is excluded.

Phase distance $d_\phi(a,b) = \min_n |\varphi_a - \varphi_b - 2\pi n|$ and metric distance d_g are independent observables. The empirical question [H] is whether $d_\phi \approx 0$ still predicts correlated records at large d_g after every standard correlated term is removed, on a named chain, under a named hypothesis, between two named input series.

3. The instrument

A DSLV-ZPDI node comprises a GNSS-disciplined oscillator, an SDR front end tracking the GPS broadcast carriers, and local housekeeping (optional E/B, barometric, or radiogenic channels may be logged; they are not estimator inputs).

Tracked carriers. GPS L1 ($f_1 = 1575.42$ MHz, $\lambda_1 \approx 19.03$ cm) and L5 ($f_2 = 1176.45$ MHz, $\lambda_2 \approx 25.48$ cm). Frequency ratio $f_1/f_2 = 154/115 \approx 1.339$. The constellation is the transmitter network; IGS final orbit/clock products are part of the null. Each node's SDR tracks every in-view satellite against the local GPSDO.

3.1 Two processing chains

An additive phase common to every tracked satellite at one antenna cancels in a between-satellite single difference. An additive phase common to one satellite at every antenna cancels

in a same-satellite inter-node difference. The title hypothesis is not a single observable. Two chains are defined and pre-registered.

Chain S — sky-gradient (H_S). Between-satellite single difference at node k, same band:

$$\varphi_k^{(pq)}(t) = \varphi_k^{(p)}(t) - \varphi_k^{(q)}(t) \quad (1)$$

cancels the receiver clock and any isotropic site-common phase at first order [E].

The between-node double difference

$$\nabla\Delta_{k\ell}^{(pq)}(t) = \varphi_k^{(pq)}(t) - \varphi_\ell^{(pq)}(t) = [\varphi_k^{(p)} - \varphi_k^{(q)}] - [\varphi_\ell^{(p)} - \varphi_\ell^{(q)}] \quad (2)$$

additionally cancels satellite-clock residuals and common-view ephemeris terms exactly — including the ≈ 75 ps IGS final clock uncertainty, which is

$$\varphi_{IGS} = 2\pi f_1 \Delta t_{clk} \approx 2\pi (1.57542 \times 10^9 \text{ Hz})(75 \times 10^{-12} \text{ s}) \approx 0.74 \text{ rad} \quad (3)$$

at L1 and, being the product's error, is identical at every node using that product [E]. The double difference is the primary Analysis-A/B input for Chain S. Chain S is blind to isotropic site-common phase. A null on Chain S bounds anomalous inter-satellite differential coherence and does not constrain H_C .

Chain C — site-common (H_C). Same-satellite inter-node single difference, same band:

$$\Delta_{k\ell}^{(p)}(t) = \varphi_k^{(p)}(t) - \varphi_\ell^{(p)}(t) \quad (4)$$

This cancels the satellite clock at first order and keeps receiver-common and site-common phase [E]. Receiver clocks do not cancel; they are bounded with IGS products, the measured ADEV of the installed GPSDOs, and the common-clock calibration (§4.4), whose transferability to distributed baselines is explicitly limited. Chain C can bound correlated site-common phase with inter-site differential. It is exposed to residual receiver-clock leakage below and near the steering bandwidth.

A licensed ground beacon remains a possible future secondary tone; it would require its own budget rows and is out of scope.

3.2 Residuals, not raw differences

Analyses A and B consume post-subtraction residuals of the named chain's estimator input: IGS final orbit and clock, tropospheric hydrostatic and wet mapping, ionospheric first-order term (bound product) or the uncombined dual-frequency pair (classification product), and the site multipath phase-error model G_{km} . Per-node single differences sharing the same IGS product carry the product's clock error in common across nodes; that coherence is the null's, not a detection — which is why the double difference, not the per-node single difference, is the Chain-S estimator input. The per-node post-IGS single difference is retained as a diagnostic that the subtraction injected no structure.

3.3 Bound product and classification product

Bound product. Iono-free or dual-frequency-corrected residual of the Chain-S double difference and, separately, of the Chain-C inter-node difference. This is the \$5 number and the

Analysis-A/B detection statistic for Switch 1. The standard iono-free combination of carrier phase (in metres) is

$$\Phi_{IF} = (f_1^2 \Phi_1 - f_2^2 \Phi_2) / (f_1^2 - f_2^2) \quad (5)$$

Classification product. L1 and L5 uncombined, same windows, same chain, same input series as any candidate. This is Switch 2. The iono-free combination mixes the two carriers and destroys the chromatic ratio; it is never the Switch 2 observable.

3.4 Timing numbers

Two timing numbers, not to be conflated:

Snapshot alignment. GNSS 1 PPS / timestamp to $\ll 100$ ns so records sit on a common grid. Bookkeeping, not the science floor. At L1, 100 ns is ≈ 157 full carrier cycles.

Phase stability. Short-term ADEV of the disciplined OCXO sets the reference contribution to $\phi_k^{(p)}$ before chain combinations. For $\sigma_y(\tau) \approx 10^{-12}$ at $\tau = 1$ s (mid-grade OCXO GPSDO [E]),

$$\sigma_\phi(\tau) \approx 2\pi f \tau \sigma_y(\tau) = 2\pi (1.57542 \times 10^9 \text{ Hz})(1 \text{ s})(10^{-12}) \approx 9.9 \text{ mrad} \quad (6)$$

A 10^{-11} unit is ten times worse. On Chain S the receiver clock cancels in the double difference; the residual requirement enters through inter-channel biases and timestamping. On Chain C the receiver clock remains and the measured ADEV is load-bearing, with the relative-clock contribution isolated by §4.4. Quote the measured ADEV of the installed unit, not a catalog class.

Synchronization requirement, per baseline. Let $\tau_c = d_g/c$. Any claim about lag structure requires $\sigma_{\text{sync}} \ll \tau_c$, where σ_{sync} is the residual uncertainty of the time-transfer method after common-view correction. For 10 km ($\tau_c = 33 \mu\text{s}$), common-view GNSS (ns-level) suffices. For co-located or sub-km baselines ($\tau_c \lesssim 1 \mu\text{s}$), White Rabbit or two-way optical transfer is required [E]. GPS 1 PPS alignment is a simultaneity convention, not a measurement; the transfer method per baseline is pre-registered (§6). An excess at “ $\tau \approx 0$ ” is only meaningful against the stated σ_{sync} of its baseline.

Guard band: where the nodes are not independent. Every GPSDO slaves its OCXO to GPS time through a steering loop of bandwidth 10^{-3} – 10^{-2} Hz. Below that bandwidth, receiver-clock leakage is common-mode by construction. Frequencies below the measured loop bandwidth of the installed units (pre-registered per unit) are excluded from both analyses on both chains. The exclusion is mandatory on Chain C and conservative on Chain S. The testable frequency band is bounded below by the steering loop and above by half the prompt dump rate; the number of independent samples within that band is bounded by the atmospheric correlation time (§4.2).

Symbol	Quantity	Unit
$\varphi_k^{(p)}(t)$	carrier phase, satellite p, vs local GPSDO (NCO + discriminator)	rad
$\varphi_k^{(pq)}(t)$	Chain S: between-satellite single difference (diagnostic)	rad
$\nabla\Delta_{k\ell}^{(pq)}$	Chain S: between-node double difference (estimator input)	rad
$\Delta_{k\ell}^{(p)}(t)$	Chain C: same-satellite inter-node difference (estimator input)	rad
f_d	prompt-correlator dump rate	Hz
$r, \hat{y}$	pairwise coherence / magnitude-squared coherence (Analysis A)	1
$\hat{C}(\tau), \hat{r}(\tau)$	lag cross-correlation / normalized lag coherence (Analysis B)	rad ² / 1
τ_k	clock offset to ensemble	s
τ_c	light time between pair-of-pairs reference points, d_g/c	s
L_{eff}	effective independent Analysis-A windows	1

Null model, correlated terms included. Thermal noise, oscillator phase noise (residual after the chain's clock cancellation), multipath, clock residuals appropriate to the chain, common-view GNSS, tropospheric hydrostatic and wet delay, ionospheric TEC gradients, and catalogued transmitters through G_{km} . Dual-frequency GNSS and IGS final orbit/clock products are part of the null [E]. Any of these produces real inter-site coherence that is not new physics.

Band selection and the $1/f$ versus $1/f^2$ argument. Ionospheric group delay (and ionospheric range on the carrier, opposite sign) scales as $1/f^2$ in metres:

$$\Delta t = 40.3 \cdot \text{TEC} / (c f^2), \quad I(m) = 40.3 \cdot \text{TEC} / f^2 \quad (7)$$

Carrier phase in radians or cycles scales as $1/f$. A 0.1 TECU uncorrected residual is ≈ 8.4 rad of differential phase at 100 MHz but ≈ 0.53 rad at L1, a factor of $\approx 15.75 = f_1/100$ MHz — not $(15.75)^2$. The dual-frequency iono-free combination removes the first-order term in the bound product [E], leaving higher-order residuals at the millimetre level. The worked budget in §5 is L-band (L1 + L5). A sub-100 MHz carrier-phase coherence measurement across km baselines is uncorrectable with current TEC products; any band other than L-band (or higher) requires its own ionospheric error row and a spectrum-survey hash before the science run.

4. Estimators

4.0 Data path: from tracked carrier to analysis streams

The tracking loop is standard GNSS practice [E] (Kaplan & Hegarty), with one correction that matters: a phase-locked loop absorbs phase dynamics inside its loop bandwidth into the numerically controlled oscillator. The per-dump observable is therefore NCO carrier phase plus the prompt discriminator residual — $\text{atan2}(Q,I)$ alone is the loop residual, not the carrier phase. Data-bit wipeoff is required on L1 C/A (50 bps); L5Q provides a dataless pilot. Cycle slips are detected and repaired by a pre-registered algorithm; unrepaired slips are flagged in the residual stream and excluded from Analysis-A windows.

1. **Raw.** Wideband complex IQ at sample rate f_s , held in a ring buffer (depth pre-registered; e.g. 300 s) for reprocessing and forensics.
2. **Tracking.** Per-satellite carrier and code tracking loops produce NCO phase plus prompt-discriminator I/Q dumps at dump rate f_d (pre-registered; nominal 1-10 kHz).
3. **Chain construction.** Form Chain-S single differences $\varphi_k^{(pq)}$ and Chain-C inter-node differences $\Delta_{kl}^{(p)}$ at rate f_d . Subtract the §3 null models. For Chain S, the double difference $\nabla\Delta_{kl}^{(pq)}$ is the estimator input: it cancels the satellite-clock residual exactly, including the ≈ 0.7 rad shared IGS term at L1. The per-node post-IGS single difference is a diagnostic of the subtraction, not a detection input — raw same-(p,q) single differences at two nodes are coherent through shared satellite terms and are never detection inputs.
4. **Analyses.** Analysis A consumes windowed spectra of the named chain's estimator input series (§4.2). Analysis B consumes the same input series in the time domain (§4.3).

Dump-rate tradeoff. Coherent integration time $T = 1/f_d$ sets both per-dump phase precision

$$\sigma_{\varphi, \text{dump}} \approx 1 / \sqrt{(2 \cdot C/N_0 \cdot T)} \quad (8)$$

and lag resolution $\delta\tau = 1/f_d$. Raising f_d sharpens the lag grid at the cost of per-dump SNR; lag-correlation integration over the trigger window recovers sensitivity per the radiometer equation. f_d is pre-registered. The Analysis-B morphology test $|\tau| \ll \tau_c$ additionally requires $\tau_c \gg f_d^{-1}$.

A wideband residual-RF lag correlation at f_s (IQ minus reconstructed known signals) is a natural Analysis-B extension with microsecond lag resolution; it requires its own subtraction-fidelity budget row and is out of scope for this revision. Until it is in scope, regional baselines do not support a spacelike morphology claim.

4.1 Residual snapshot

On the post-subtraction residual of a named chain's estimator input, the analysis streams are the windowed Fourier coefficients $\tilde{y}_A, \tilde{y}_B$ of the two named input series (Analysis A) or the same series in the time domain (Analysis B). Chain index (S or C), product (bound or classification), and the pair-of-pairs construction are part of the series name in the archive.

4.2 Analysis A — stationary coherence (bound)

Named inputs. Coherence is computed between two input series on disjoint node pairs:

- **Chain S:** double-difference series $\nabla\Delta_{k\ell}^{(pq)}$ and $\nabla\Delta_{mn}^{(pq)}$, $\{k,\ell\} \cap \{m,n\} = \emptyset$, same satellite pair and band. All clock terms cancel exactly in each series; under the null the two series are independent.
- **Chain C:** inter-node difference series $\Delta_{k\ell}^{(p)}$ and $\Delta_{mn}^{(p)}$, $\{k,\ell\} \cap \{m,n\} = \emptyset$, same satellite. Receiver clocks are independent across the four nodes; under the null the two series are independent, and under H_C they share the correlated site-common differential.

On L non-overlapped windows the magnitude-squared coherence is

$$\hat{\gamma} = |\hat{S}_{AB}|^2 / (\hat{S}_{AA} \hat{S}_{BB}) = (\hat{r}^{(res)})^2 \quad (9)$$

For windowed spectral coefficients that are complex-Gaussian under the null (asymptotic for stationary series; approximate for 1/f-type residuals — hence the bootstrap below), $\hat{\gamma} \sim \text{Beta}(1, L-1)$ [E] (Goodman; Carter; Gish and Cochran invariance), so

$$E[\hat{\gamma}]_{\text{null}} = 1/L, \quad P(\hat{\gamma} > \gamma \mid H_0) = (1 - \gamma)^{L-1} \quad (10)$$

Two qualifications: (i) the Beta law assumes non-overlapped windows and nominal degrees of freedom; overlapped windows alter the DOF and are not used in the science run; (ii) per-window re-phasing $e^{i\omega\hat{\tau}_k}$ consumes degrees of freedom and perturbs the null at $O(1/L^2)$; the block bootstrap, not the analytic tail, is the decision statistic.

Independence is not free. Calendar seconds are not independent. Multipath dwell, flicker frequency noise, GNSS geometry (≈ 11.97 h repeat), and diurnal ionosphere set a correlation time τ_{corr} . Oscillator and atmospheric residuals are 1/f-type processes, for which a lag-1 autocorrelation is ≈ 1 at almost any sampling and carries almost no information about the effective sample size. Accordingly

$$L_{\text{eff}} = T / \tau_{\text{corr}} \quad (11)$$

with τ_{corr} obtained two ways on pilot data and reconciled before the science run: (a) the Politis-White automatic block-length selection for the moving-block bootstrap applied to the input series [E]; (b) an Allan-deviation-informed independence time from the measured $\sigma_y(\tau)$ turnover of the installed units [E]. The block length is then frozen in the pre-registration (§6). Detection uses L_{eff} or the bootstrap, never raw L .

Cramér-Rao phase floor on a tone in complex AWGN, N_s samples [E] (Kay 1993):

$$\sigma_{\phi, \text{CRB}} \approx 1 / \sqrt{(2 \cdot \text{SNR} \cdot N_s)} \quad (12)$$

Array size. Detection-grade Analysis A requires two disjoint pairs: ≥ 4 nodes. A 2-3 node deployment executes the Phase-0 validation campaign (§6) — floor measurement, injection recovery, null characterization — and may contribute single-baseline Analysis-B lag classification, but no Analysis-A detection statistic.

Multiplicity. Pair-of-pairs test statistics sharing nodes are positively correlated; bands and the two chains are likewise correlated through common systematics. Benjamini-Hochberg

FDR control remains valid under positive regression dependence (PRDS) [E]. Chain S and Chain C are separate families in the pre-registration; FDR is applied within family and the joint accounting is stated.

4.3 Analysis B — lag-resolved cross-correlation

Analysis B consumes the same two named input series as Analysis A, in the time domain at dump rate f_d , lag resolution $\delta\tau = f_d^{-1}$:

$$\hat{C}_{AB}(\tau) = \langle y_A(t) y_B^*(t - \tau) \rangle \quad (13)$$

Light-travel time between the pair-of-pairs reference points (e.g. pair centroids, pre-registered): $\tau_c = D/c$.

Resolution requirement. The morphology test $|\tau| \ll \tau_c$ requires $\tau_c \gg f_d^{-1}$. Baselines not satisfying this at the pre-registered f_d are resolution-limited for the lag test: they contribute fully to Analysis A and to the coarse Analysis-B grid; no $|\tau| \ll \tau_c$ claim is made on them.

f_d	$\delta\tau = f_d^{-1}$	D at $\tau_c = \delta\tau$	D at $\tau_c = 10 \delta\tau$
1 kHz	1 ms	300 km	3000 km
10 kHz	100 μ s	30 km	300 km
IQ @ 2 MHz (deferred)	0.5 μ s	150 m	1.5 km

At nominal dump rates, a 10 km pair ($\tau_c = 33 \mu$ s) and a 100 km pair ($\tau_c = 333 \mu$ s) are resolution-limited for $|\tau| \ll \tau_c$. Analysis B on those baselines is a coarse-lag companion to Analysis A, not a spacelike morphology test. Raising f_d or activating the deferred IQ path extends the testable set; that choice is pre-registered, not tuned post hoc.

A 1 s Analysis-A window cannot locate a peak to better than 1 s and cannot declare a lag spacelike on any terrestrial baseline. Analysis B publishes $\hat{r}_{AB}(\tau)$ on a pre-registered τ grid containing $\{0, \pm\tau_c, \pm\tau_{iono}, \pm\tau_{trop}\}$ and a uniform span of $\pm M \tau_c$ (M pre-registered), at native resolution $\delta\tau = f_d^{-1}$.

Multiplicity control. The detection statistic is the maximum of $\hat{r}_{AB}(\tau)$ over the grid. Significance is set by a moving-block bootstrap of that maximum (block length per §4.2). No per-lag κ claims are permitted.

Trigger-selection bias and its closure. Deep Analysis-B follow-up runs on windows selected because Analysis A crossed threshold — the same noise realization. The pipeline is calibrated end-to-end: the full chain (A selection $\rightarrow$ B on selected windows) is run on time-slides — one input series shifted by unphysical offsets $\gg \tau_{corr}$ — with the slide count and the joint false-alarm rate pre-registered [E]. Slides are applied to the post-subtraction residual streams. Correlated residuals that survive the §3 subtraction are not part of the slide background; they are the subject of Switch 1. A candidate must additionally survive a hold-out confirmation window: post-window residual data, excluded from the detection statistic, must independently show the morphology at the pre-registered level.

End-to-end validation. Before any detection claim, blind injections are run into realistic residuals — nonstationary, with data gaps, geometry changes, and clock steering — including null-true cases with an imperfect noise model. The program reports empirical false-positive rates and coverage of the resulting bounds, not just that the code implements the prescribed tests.

Triggered architecture. Analysis A runs continuously on each chain input. Each node maintains the wideband-IQ ring buffer for reprocessing; when Analysis A exceeds its pre-registered trigger threshold in any band, the buffer ($\pm$ window) is dumped at all nodes simultaneously. Trigger-distribution latency ($\ll 1$ s over any terrestrial IP path) is negligible against the 300 s dwell. Continuous full-rate B recording is not required. Chain residuals are continuous and cheap (tens of kB/s per node).

Interpretation, pre-registered, per chain:

- Direct propagation from any single source satisfies $|\tau| \leq \tau_c$, with equality only for a source on the inter-pair axis. Peak at $\tau = \tau_c$ or a catalogued multipath extra delay: ordinary propagation. Switch 1 fails for that candidate.
- A peak at $|\tau| > \tau_c$ cannot be direct propagation. If it matches no catalogued multipath delay, it is dispersive or instrumental and triggers a pipeline investigation under Switch 3.
- Peak at $|\tau| \ll \tau_c$ (including $\tau \approx 0$ within the stated σ_{sync}), on a pair-of-pairs satisfying $\tau_c \gg f_d^{-1}$, that vanishes in the co-located control and vanishes when IGS/iono terms are applied: a candidate for injection-matched follow-up, subject to Switch 2, labelled with its chain and hypothesis.
- Peak at $|\tau| \ll \tau_c$ that also appears in the co-located pair-of-pairs is site-common hardware unless Chain C is the intended target and the common-clock calibration has excluded the hardware reading. Co-located excess is not anomalous on Chain S.
- A common mode that is real on Chain C (H_C) would appear at $|\tau| \ll \tau_c$ in both co-located and separated pair-of-pairs, with $\delta\tau \ll \tau_c$ and $\sigma_{\text{sync}} \ll \tau_c$, would survive the common-clock decomposition in §4.4, and would pass Switch 2 in the delay class. That is the only Chain-C morphology worth a data paper, and only on baselines that are not resolution-limited.
- A $\tau \approx 0$ excess, even if every switch is passed, is common-mode noise. It is not superluminal signaling. No information transfer is possible via a common-mode field.

4.4 Injection / recovery and common-clock calibration

Before any science run: inject synthetic correlated phase at known amplitude and known lag into the raw stream; run the full pipeline on both chains; require recovery of amplitude and lag within pre-registered tolerance; require return to the null floor with injection off. Injections include (i) correlated site-common differential phase (must appear on Chain C and vanish on Chain S — the H_C signature) and (ii) direction-differential phase (must appear on Chain S — the H_S signature). A chain that fails its distinctive injection has no standing to report an excess.

Blind injections. A pre-registered number of injections per campaign are seeded by an injection holder who is not on the analysis team, with amplitude, lag, and chain-class drawn

from pre-registered distributions and revealed only after the analysis of record is frozen. Open injections test the pipeline; blind injections test the analysts. Small-team implementation: a sealed-seed script held by a designated non-analyst; the mechanism is pre-registered, not the seed.

Common-clock calibration (breaks the Chain-C clock degeneracy). Before the science run, and repeated per campaign: one GPSDO output split to drive two complete front-end + SDR chains at a single site (common clock; independent RF paths and antennas at metre-scale separation). Chain C formed between these two front ends contains site-common phase and inter-channel biases but no relative clock wander by construction — it measures the non-clock systematic floor of Chain C directly, including per-channel L1/L5 inter-channel bias differences (the phase-offset-class confound for Switch 2). The independent-clock co-located pair measures site-common plus relative clock wander; differencing the two co-located configurations isolates the relative-clock contribution:

$$\sigma_{\text{clk}} = \sqrt{(\sigma_{\text{indep}}^2 - \sigma_{\text{common}}^2)} \quad (\text{require } \sigma_{\text{indep}} \geq \sigma_{\text{common}}) \quad (14)$$

If the common-clock run is noisier than the independent-clock run, the decomposition has failed and Switch 3 fires. A common-clock pair that does not sit on its predicted null routes to Switch 3 before any science claim.

Transferability. The calibration characterizes one site’s hardware; it does not automatically describe separated sites.

Term	Measured directly?	Treatment
Inter-channel (L1/L5) phase biases	Yes	per-band, per-campaign; bounds the Switch-2 phase-offset confound
Front-end + splitter phase and thermal drift	Yes	at the calibration site’s thermal environment
Co-located multipath / RFI / supply	Yes	site-specific; not transferable to other sites
Relative receiver-clock wander	Yes, by Eq. (14)	enters the Chain-C max bound
Baseline-dependent tropospheric differential	No	decorrelates with distance; budgeted per baseline from §5
Far-site multipath decorrelation	No	per-site G_{km} ; budgeted per baseline
Independent GPSDO drift between sites	No	bounded by per-unit ADEV plus interleaved spot checks

The calibration repeats per campaign with interleaved spot checks during long runs; measured drift variance between repeats is added in quadrature to the Chain-C floor. A spot check deviating beyond the pre-registered tolerance routes to Switch 3. Per baseline, the reported Chain-C bound is

$$B_C = \max(\sigma_{\text{common}}, \sigma_{\text{clk}}, \sigma_{\text{atm,baseline}}) \quad (15)$$

No Chain-C number is quoted tighter than that maximum.

5. Sensitivity budget (worked example, Analysis A, L-band)

Order-of-magnitude engineering estimates, except where marked measured. Differential noise sets the per-snapshot floor σ_φ ; correlated threats must be subtracted by the null, and whatever survives is adjudicated by Switch 1. Numbers below are for the bound product. Every results table in the program reports three columns — predicted budget (this section) | measured residual floor (validation campaign) | reported bound (pre-registered analysis after systematics). Never let the first imply the third.

(a) Differential noise rows → per-snapshot σ_φ (RSS)

Parameter	Assumed value	Basis
Carriers	GPS L1 1575.42 MHz + L5 1176.45 MHz	\$3
Chain S observable	between-node DD of per-node post-IGS SD	satellite and receiver clocks cancel exactly [E]
Chain C observable	same-satellite inter-node SD residual + IGS	satellite clock cancels; receiver clock remains [E]
Tropospheric wet-delay residual	1 cm $\Rightarrow \approx 0.33$ rad at L1	Saastamoinen-class mapping limits [E]
Multipath residual, post-model	0.5–2 cm $\Rightarrow 0.17$ –0.66 rad	site-dependent specular bounce [E]
Antenna PCV + cable + front-end	placeholder 0.1 rad; measured per unit	catalog values not accepted
Thermal noise on carrier	1 mrad	high-SNR broadcast carriers
GPSDO reference	10 mrad at 1 s before combination	installed ADEV; cancels on Chain S
Shared IGS clock residual	≈ 0.7 rad at L1 (75 ps class)	cancels exactly in Chain-S DD; would dominate per-node SD coherence
Chain-C clock terms	measured, per campaign	common-clock calibration; enter the per-baseline max, not the RSS
Per-snapshot σ_φ (RSS), Chain-S DD	≈ 0.4 –1.1 rad	tropo + multipath, $\sqrt{2}$ over single difference

(b) Correlated-threat rows → null subtraction + Switch 1 (not in the RSS)

Threat	Handling
Common-view satellite orbit/clock residuals (IGS final: orbit ≈ 2 cm, clock ≈ 75 ps)	cancelled exactly in the Chain-S double difference [E]; subtracted per-node via IGS on Chain C [E]; survival $\rightarrow$ Switch 1
Ionospheric gradients (first order removed in the bound product)	dual-frequency combination [E]; higher-order residual characterized on pilot data; survival $\rightarrow$ Switch 1
Site-common RFI / supply / local field	co-located pair-of-pairs. On Chain S this class must vanish; on Chain C it is the hypothesis class (H_C) and is separated from hardware by the common-clock calibration, Switch 2, and Switch 6
Inter-channel (L1/L5) bias differences	calibrated per campaign in the common-clock run; the phase-offset-class confound for Switch 2
Transmitter catalog leakage	catalog version frozen; injection bounds pipeline leakage

Run parameters (illustrative, not a result). Calendar snapshots: 86,400 (24 h at 1 s Analysis-A windows). L_{eff} : 10^2 – 10^3 on pilot data. Detection: block-bootstrap null on $\hat{y}$, FDR (PRDS) per chain family; bootstrap-max over τ grid for B; joint FAR by time-slides on residuals; empirical FAR from blind end-to-end simulations.

If one incorrectly used $L = 86,400$, the Beta null would quote $E[\hat{y}] = 1/86400 \approx 1.2 \times 10^{-5}$. That number is not the program floor. For $L_{\text{eff}} = 300$ ($\tau_{\text{corr}} = 288$ s): $E[\hat{y}]_{\text{null}} = 1/300 \approx 3.3 \times 10^{-3}$, $r_{\text{min}} \sim O(10^{-1})$. The statistical floor moves as $\sqrt{(86400/L_{\text{eff}})}$. Systematics sit above both.

Weak common phase. Two input series, shared phase φ_s (rms), independent residual phase σ_φ , treating the phase records as the measured quantity:

$$r \approx \varphi_s^2 / (\varphi_s^2 + \sigma_\varphi^2) \quad \Rightarrow \quad \varphi_s \approx \sigma_\varphi \sqrt{r} \quad \text{when } \varphi_s \ll \sigma_\varphi \quad (16)$$

The DD floor $\sigma_\varphi \approx 0.4$ – 1.1 rad is not $\ll 1$; the millimetre conversion is an order-of-magnitude map, not an estimator. With $\sigma_\varphi = 0.42$ rad: calendar $r_{\text{min}} = 7.7 \times 10^{-3}$ gives $\varphi_s \approx 37$ mrad, ≈ 1.1 mm equivalent path at L1. With $L_{\text{eff}} = 300$: $\varphi_s \approx 0.13$ rad, ≈ 4.0 mm. Report both, report the measured L_{eff} , and report Chain S and Chain C separately. The systematic floor governs.

Deliverable. Two bounds, set by different floors. Chain S (H_S): upper bound on anomalous sky-gradient residual phase at $O(0.1)$ rad) (mm-cm equivalent path at L1) under standard geodetic site engineering, improving only as the measured rows improve. Chain C (H_C): per-baseline upper bound on correlated site-common differential phase, set by Eq. (15). Multipath and troposphere decorrelate between nodes and inflate σ_φ rather than fake a signal; correlated terms are the danger and are handled by subtraction plus Switch 1.

6. Pre-registration and data release

Frozen before first science run, timestamped public registry. Phase 0 is a validation campaign: publish the achieved residual floor, measured L_{eff} , injection-recovery report, and

empirical false-alarm behavior before any search claim. No detection statistic from a science run is reported without the Phase-0 null characterization on record.

- science band and spectrum-survey hash
- signal definition: tracked signals, NCO+discriminator observable, data-bit wipeoff method, cycle-slip algorithm, f_d , elevation masks, per-dump SNR characterization
- chain definitions and distinctive injection waveforms for H_S and H_C
- product definitions: bound product vs classification product
- Analysis A: window length (non-overlapped), Politis-White parameters and frozen block length, Allan cross-check, guard band f_{loop} per installed unit, selection thresholds to B, FDR procedure with PRDS clause and per-chain families, transmitter catalog version, G_{km} version per site, minimum array (≥ 4 nodes) for detection-grade analysis
- Analysis B: τ grid (M, special lags), pair-of-pairs reference-point convention and per-pair τ_c , bootstrap-max procedure, peak-location classifier, time-slide count and joint FAR target, hold-out window criteria, resolution-limited baseline list at the chosen f_d
- end-to-end simulation report: blind injections into realistic residuals, empirical false-positive rates, coverage of reported bounds
- IQ ring-buffer depth, dump window, multi-node dump coordination
- per-baseline time-transfer method and stated σ_{sync}
- open injection schedule; blind-injection count, distributions, and the identity (not the data) of the injection holder
- common-clock calibration: splitter and front-end configuration, run schedule and interleaved spot checks, transferability table, drift-variance propagation, predicted-null pass/fail criteria, per-band inter-channel bias calibration
- file hashes; installed GPSDO ADEV curves; measured per-node analog-chain phase vs temperature; antenna PCV calibration file hashes; co-located-pair mutual-coupling characterization
- Switch 2 pass criteria: chromatic classes with uncertainty intervals on the classification product; multi-component mixture policy
- exact null including §3 correlated terms, per chain, per hypothesis, per named input pair
- three-column results schema: predicted budget | measured residual floor | reported bound

Freeze anchoring. The registry freeze hash (SHA-256 over the canonical registry) is anchored externally before the science run: an OSF preregistration, a signed git tag in the public DSLV-ZPDI repository, or an OpenTimestamps / RFC-3161 timestamp. The anchor transaction ID is part of the registry. A hash stamped only by the handset clock is self-attested; without an external anchor the freeze is a draft, not a pre-registration. The field-book implementation (Appendix A) prompts for export (JSON archive) at every frozen milestone — registry freeze, amendment re-freeze, calibration append, campaign close — not only at publication.

Raw IQ or residuals, pipeline code, injection logs (open and, after unblinding, blind), slide backgrounds, simulation reports, and the frozen pre-registration are published regardless of

outcome. A null is a publishable pair of upper bounds (Chain S $\rightarrow$ H_S, Chain C $\rightarrow$ H_C) at the measured L_{eff} and systematic floor.

7. Kill switches

Instrument switches only. Four switches are pre-registered. Former ontology switches from earlier drafts are withheld with Supplements S1/S2 and are not part of this submission.

Switch 1 — Array residual (primary), per chain, per hypothesis

Fires on: Analysis A $\hat{y}$ on the disjoint-pair input series consistent with the Beta / bootstrap null after pre-registered subtraction, FDR-controlled (PRDS), every pair-of-pairs and band, separately for Chain S (H_S) and Chain C (H_C). Analysis B: the bootstrap-max over the τ grid is consistent with the time-slide-calibrated null; any excess peaks at $|\tau| \geq \tau_c$ (noting that $|\tau| > \tau_c$ without a catalogued multipath explanation routes to Switch 3), vanishes when common-view/ionospheric/IGS terms are included, or — on Chain S — is confined to the co-located control. The anomalous morphology class $|\tau| \ll \tau_c$ is claimed only on pair-of-pairs with $\tau_c \gg f_d^{-1}$. A null on Chain S is not a null on Chain C.

Blast radius: the hypothesis as tested by that chain. The array remains a disciplined SDR network.

Switch 2 — Chromaticity (classification product only)

GPS L1 and L5 are separated by $f_1/f_2 \approx 1.339$. Let φ be the candidate residual in radians or cycles, not metres. Three classes, applied with uncertainty intervals to the uncombined L1/L5 pair:

Class	φ_1/φ_2	Reading
Delay	$f_1/f_2 \approx 1.339$	equal in seconds (clock, time-transfer) or metres (path). The only class that may proceed.
Ionospheric	$f_2/f_1 \approx 0.747$	carrier-phase ionospheric residual in radians/cycles scales as $1/f$. Routes to the null as ionosphere.
Phase-offset	1	equal in radians. Digital/processing artifact. Routes to instrumental investigation.

Passing Switch 2 is necessary, not sufficient: tropospheric delay, multipath, and clock error are all delay-class, so a delay-class ratio rejects only ionospheric and digital-artifact explanations. Real residuals are mixtures; the classifier is multi-component; hardware frequency dependence is bounded by the common-clock calibration; a candidate matching no single class is not promoted on chromaticity. The iono-free bound product is not used here. Blast radius: the candidate, not the array. This switch cannot fire on a null.

Switch 3 — Pipeline non-recovery

Fires if open or blind injection of known correlated phase at known lag and known chain-class is not recovered within the pre-registered window (including the distinctive S-vs-C pattern); or the off-injection state does not return to the null floor; or the time-slide background

disagrees with the bootstrap null at the pre-registered level; or the end-to-end simulation's empirical FAR disagrees with the reported FAR; or the common-clock calibration pair (or an interleaved spot check) does not sit on its predicted null. Blast radius: the campaign, until the pipeline is repaired. No science claim is permitted.

Switch 6 — Simultaneity convention

If an Analysis-B $|\tau| \ll \tau_c$ excess is unstable under substitution of an independent time transfer (two-way optical, common-view vs all-in-view, or a second GNSS constellation), or does not survive substitution of the calibration configuration (common-clock spot check), the excess is a clock-ensemble or hardware artifact. Blast radius: any non-local reading of that dataset. Load-bearing on Chain C (H_C).

Numbering keeps Switch 6 as in Rev 3.5 so field-book and repository labels do not fork. Switches 4 and 5 are not used in this compilation.

8. Closing

Delayed light is still light. A correlation is a subtraction artifact until the null model — including every known correlated term and the measured correlation time — fails to contain it. GPS alignment is a convention. A 1 s average is not a spacelike sample on Earth. Below the steering bandwidth of a disciplined oscillator, two GPSDOs are one clock. A between-satellite single difference is blind to isotropic site phase. A dump-rate lag axis that is coarser than d_g/c cannot test $|\tau| \ll \tau_c$. An isotropic common phase is indistinguishable from relative clock wander until a common clock says otherwise — and the common clock only speaks for its own site. A predicted budget is not a measured floor, and a measured floor is not a reported bound.

This program tests no predicted amplitude and specifies no coupling model. It sets two upper bounds on anomalous distributed RF phase coherence — Chain S on the sky gradient (H_S), Chain C on correlated site-common differential phase (H_C) — at a measured L_{eff} and measured systematic floors, reported as predicted | measured | bound. A null is the expected result. “Systematics-limited” describes the bound, not a failure.

Run Phase 0 first: publish the achieved floor, the measured L_{eff} , the injection recoveries, and the empirical false-alarm behavior before any search claim. Build clocks good enough that phase is a spatial field. Name the chain, the hypothesis, and the two input series. Cancel the clocks each chain is allowed to cancel — exactly, in the double difference, not approximately in a product. Subtract Helmholtz, multipath, the troposphere, IGS, and the ionosphere. Reconstruct the carrier from NCO plus discriminator; wipe the data bits; flag the slips. Define the data path from tracked carrier to estimator before the first science run. Measure τ_{corr} with an estimator that respects $1/f$ noise. State σ_{sync} and $\delta\tau$ per pair-of-pairs. Split one clock across two front ends, measure Chain C's non-clock floor, bound its transferability, and take the max. Anchor the freeze outside the handset. Slide the residuals against themselves until the background is measured. Inject known lags on both chain-classes — some of them blind, into realistic residuals — and prove the pipeline finds them, puts them on the correct chain, and reports an honest false-alarm rate. Publish residuals, $\hat{r}^{(\text{res})}$, and $\hat{r}(\tau)$. If they sit on the null, the bound is the result. If they do not, the candidate still has to survive chromaticity on

the uncombined pair (necessary, not sufficient), an independent clock, a hold-out window, and a quantitative signal model that no conventional mechanism reproduces — and only then is the next paper a data paper, not another ontology.

Data availability

All raw IQ or residual streams, pipeline code, injection logs (open and blind, the latter after unblinding), time-slide backgrounds, simulation reports, and the frozen pre-registration will be published regardless of outcome, per §6. No part of the analysis of record may be tuned on the science data after the registry timestamp.

Competing interests

The author declares no competing financial interests. Resonant Genesis LLC funds the instrument program.

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DSL-V-ZPDI Field Book

Rev 3.6 Install and User Guide

Phone-ready campaign record for GrapheneOS on a Pixel 9 Pro XL. The handset keeps the campaign record. The SDR, GPS-disciplined oscillator, and prompt correlator remain on the DSL-V-ZPDI nodes.

Install address: <https://dslv-vac-probe.grok.me>

Read this first

- The field book does not make a measurement. It does not synthesize IQ, residuals, magnitude-squared coherence, or a detection. It records what the node reduction produced and enforces the Rev 3.6 gates.
- No Play Store package. Install the published web app directly from its HTTPS address.
- No account. The campaign book lives in on-device site storage in the browser profile where you install it.
- No cloud copy. Export at every frozen milestone and before clearing site data, changing origins, or removing a campaign.
- A local hash is not a pre-registration. The SHA-256 must also be externally anchored and the anchor transaction ID recorded.
- A null is a valid result. Chain S and Chain C produce separate bounds under separate hypotheses.

A.1 Install on GrapheneOS

1. On the Pixel, open Vanadium. It is GrapheneOS's browser and does not require Play Services.
2. Use a normal tab, not Incognito. Open <https://dslv-vac-probe.grok.me>.
3. Confirm the padlock and the exact host dslv-vac-probe.grok.me.
4. Tap the three-dot menu at top right.
5. Tap Install app if offered. If the menu instead says Add to home screen, use that.
6. Confirm the name, place the icon, then open the field book from the home screen.

If Add to home screen seems to do nothing: press and hold the shortcut icon in the placement sheet and drag it onto the home screen. This works around a Vanadium quirk seen on some builds.

Stock Android / Chrome. Chrome on a stock Pixel uses the same menu choices. The book remains local to that Chrome profile. Do not install it in Incognito. A chat preview or another hostname is a different site and a different book.

Permissions worth setting once

Control	Set it this way
Network	Allow Vanadium network access. Without it, the site cannot load.
Location	Optional. Needed only for Capture a device fix. You may deny it and enter surveyed coordinates.
JIT	Leave off unless a loaded screen stays blank; then allow JIT for this origin only in Vanadium site settings.
Storage	Do not clear site data for this origin until you have exported. There is no cloud copy.
Updates	Open the published address again. The home-screen icon continues to point to that origin.

Location is not carrier phase. A handset fix is a GNSS position with an accuracy estimate and timestamp. It is not a monument, a residual, or a carrier-phase measurement. The fix is not uploaded by this app. Grant or deny Location under the Pixel app permission for Vanadium; if granted, answer the site prompt for this origin; read the fix, accuracy, time, and source label before saving.

A.2 Five screens, one campaign book

Screen	Address	Job
Deck	/	Open a campaign; review freeze, roster, and formula identity.
Walk	/walk	Run the thirteen gated steps.
Bench	/bench	Closed-form calculators; no automatic log writes.
Book	/book	Notes, log, export, import, and delete.
Switches	/doctrine	Kill switches, epistemic labels, and handset limits.

The five tabs sit at the bottom of the screen. After a record exists, the campaign picker appears in the header. A different hostname uses different site storage and therefore opens a different local book.

Open a campaign

1. Name the campaign. Campaign name is required. Operator and site may be blank, but they belong in the registry.
2. Tap Create the record. The record is stored only on this handset.
3. Replace catalog budget rows with measured values when available. Catalog rows remain explicitly labeled until replaced.
4. Use the header menu to switch campaigns. One handset can hold several; they do not sync.
5. Read the deck status: node count, log count, completed pre-registration fields, and whether a common-clock floor is present.
6. Use Continue the walk to return to the step you left.

Formula identity is an app self-check. The deck compares closed forms with the registered Rev 3.6 identities. A miss is an app bug, not a residual or measurement. Do not interpret it as physics.

A.3 The Walk is gated

The top bar can jump between steps, but Next remains disabled until the current gate is satisfied. Back is always allowed. Read the plate and paragraphs, then enter only what you actually did. A silent skip is not allowed; “no measurement this session” is an honest record.

Step	Gate to leave it
1, 2, 7	Tick “I have read this step against the paper.”
3	At least two fully identified nodes for Phase 0; four nodes for detection-grade disjoint-pair analysis.
4	All pre-registration fields except calibration floor and bias hash.
5	Calibration results, transferability record, and bias-file hash; export prompted after append.
6	At least one injection log: recovered, missed, or sealed.
8	SHA-256 frozen, externally anchored, anchor ID recorded, and JSON exported.
9-12	Frozen and externally anchored registry plus acknowledgement. Empty science log is allowed.
13	Campaign-close export is required from Book.

Step 1. Name the hypotheses

H_S / Chain S. Estimator input: between-node double difference of between-satellite single differences. Blind to isotropic site phase. A null on Chain S does not constrain H_C .

H_C / Chain C. Estimator input: same-satellite inter-node difference. A phase literally identical at every antenna cancels and is unobservable; H_C is the correlated site-common term that differs between sites and survives the difference.

Expected outcome: a null. The bound is the product; no amplitude or coupling model is assumed.

Step 2. From carrier to estimator

- Raw IQ stays on nodes. The phone does not hold the wideband ring buffer.
- Tracking records the full carrier observable: NCO carrier phase plus prompt-discriminator residual at the registered dump rate f_d . $\text{atan2}(Q,I)$ alone omits in-band phase absorbed by the loop.
- Apply L1 data-bit wipeoff and the frozen cycle-slip detection, repair, and exclusion procedure.
- Chain S uses the double difference. It cancels receiver clocks and the shared satellite-clock product error exactly. Per-node post-IGS single differences are diagnostics, not detection inputs.
- Chain C keeps receiver clock. Below the GPSDO steering bandwidth, nodes are one clock; exclude those frequencies on both chains.

Analyses A and B consume two disjoint node-pair series. The chain, satellite or satellite pair, band, and pair-of-pairs identifiers belong in the archive.

Step 3. Put the array on the page

Field	Record
Name	Stable node identifier.
GPSDO identity	Installed unit and serial, not a catalog class.
Measured σ_y at 1 s	Unitless Allan deviation of this installed unit.
f_{loop}	Measured steering-loop bandwidth in hertz.
Antenna	Installed antenna identity.
Latitude / longitude	Surveyed coordinate or labeled device fix.

The page computes σ_ϕ at L1 and $\tau = 1$ s as $2\pi f \tau \sigma_y$. Chain S cancels this clock term. Chain C does not. No coordinate means no baseline and no later lag claim. Two identified nodes form one pair for Phase-0 validation; four are the detection-grade minimum for two disjoint pairs.

Step 4. Write the registry

Field	What belongs there
Science band	L1 + L5. Another band needs its own ionospheric row and spectrum-survey hash.
Spectrum hash	The real SHA-256; the app does not invent one.
Dump rate f_d	Hertz; nominally 1-10 kHz; fixed before freeze.
Elevation mask	Degrees; excluded satellites never enter a chain.
Analysis A window	Seconds; non-overlapped.
Block length	Frozen Politis-White value from pilot data, reconciled with Allan turnover.
Allan cross-check	Measured $\sigma_y(\tau)$ turnover of installed units.
FDR procedure	Benjamini-Hochberg under PRDS; S and C are separate families.
Catalog / G_{km}	Transmitter catalog version and site multipath-model version.
Named inputs	For each chain, the two disjoint node-pair series used by Analyses A and B.
Analysis B grid	Half-width M in units of τ_c ; include 0, $\pm\tau_c$, $\pm\tau_{iono}$, $\pm\tau_{trop}$, and $\pm M \tau_c$.
Slides and FAR	Time-slide count and joint false-alarm rate on post-subtraction residual streams.
Hold-out	Post-window data excluded from the detection statistic.
IQ buffer	Depth in seconds; example 300 s. Trigger dumps coordinate across nodes; IQ stays on nodes.
Time transfer	Method and σ_{sync} per baseline class.
Open injections	Amplitudes, lags, and chain class: correlated site-common differential for C, direction-differential for S.
Blind injections	Count and distributions; register mechanism, never the seed. Identity of a non-analyst holder.
Switch 2 tolerance	Relative tolerance and multi-component mixture policy for ϕ_1/ϕ_2 ; example 0.05.
Phase-0 validation	Residual floor, measured L_{eff} , injection recovery, empirical FAR, and bound coverage before any search claim.
Calibration transfer	Repeat schedule, interleaved spot checks, drift tolerance, and per-baseline atmospheric treatment.

Leave calibration values for Step 5. A two- or three-node campaign validates the instrument and null. Detection-grade Analysis A begins only after four ready nodes and two disjoint node-pair inputs are frozen. At 10 km, common-view GNSS may be suitable. At sub-kilometre scale, register White Rabbit or two-way optical transfer. In every case, state the achieved σ_{sync} .

Step 5. Split one clock and bound transfer

One GPSDO output drives two complete front ends at one site, with independent RF paths and antennas about a metre apart.

1. Record the common-clock residual RMS in radians. It measures the co-located non-clock floor.
2. Record the independent-clock co-located RMS. When the independent value is larger, isolate the relative-clock term by Eq. (14).
3. Record the L1/L5 inter-channel bias file hash, thermal conditions, and scheduled interleaved spot checks.
4. Describe what the run actually did, then append the calibration and immediately export the campaign JSON.

If the common-clock run is noisier than the independent-clock run, the decomposition failed and Switch 3 fires. No science claim is permitted. Add measured drift variance between repeat runs in quadrature. The Chain C bound is Eq. (15). Never quote a tighter number.

Step 6. Prove the pipeline

Injection class	Required signature
Isotropic site-common differential	Appears on Chain C and vanishes on Chain S.
Direction-differential	Appears on Chain S in the registered form.
Injection off	Returns to the null floor.

Recovered: amplitude, lag, and chain pattern land inside the registered tolerance. Not recovered: Switch 3 is live. Still sealed: tick the sealed state and do not enter an amplitude. At least one injection line — recovered, missed, or still sealed — must be in the book before leaving this step.

Step 7. Mark resolution limits

For each surveyed pair, combine baseline distance with f_d . Light time is $\tau_c = d/c$; native lag resolution is one dump sample. No coordinates: no baseline and no lag claim. Resolution-limited pairs may still enter Analysis A, but they do not get a spacelike-morphology sentence. The claim $|\tau| \ll \tau_c$ requires $\tau_c \gg 1/f_d$ and $\sigma_{\text{sync}} \ll \tau_c$. Raising the dump rate after freeze is not a valid repair.

Step 8. Freeze and anchor

1. Freeze only when the registry, required nodes, calibration fields, and bias hash are present.
2. Let the app hash the canonical registry with SHA-256 and stamp the handset time. Treat this local stamp as self-attested.
3. Anchor the hash through an OSF preregistration, a signed git tag in the public DSLV-ZPDI repository, or an OpenTimestamps / RFC-3161 timestamp.
4. Record the external anchor transaction ID, then export the campaign JSON immediately.
5. After freeze, science logs are append-only; node and registry edits lock.

Amendments create a new commitment: give the reason, preserve the prior digest, re-freeze, create a fresh external anchor, record its transaction ID, and export again before new science entries.

Step 9. Enter Analysis A

You may draft the arithmetic before freeze. Logging stays disabled until the registry is frozen. Enter chain and named inputs, measured $\hat{y}$ from the node reduction, duration T , and τ_{corr} . The page computes $L_{\text{eff}} = T/\tau_{\text{corr}}$, $E[\hat{y}] = 1/L_{\text{eff}}$, and the ratio $\hat{y} / E[\hat{y}]$. The ratio is not a detection threshold. Calendar seconds are not independent. The large-L floor using $L = 86,400$ is the paper's trap, not the program floor. Call the entry consistent with the null, excess held for switches, or no measurement this session.

Step 10. Classify a measured lag

Required: pair, chain, peak lag τ , and σ_{sync} . Set only true flags: seen on co-located pair; seen on separated pair; catalogued multipath match; vanishes with IGS or ionosphere correction; common-clock run excluded hardware; Switch 2 delay class passed. A peak at about zero lag, even if every switch is passed, is common-mode noise. It is not superluminal signaling. Ordinary direct propagation obeys $|\tau| \leq \tau_c$. A peak beyond τ_c without a catalogued multipath explanation routes to Switch 3.

Step 11. Run Switch 2

Enter ϕ_1 for L1 and ϕ_2 for L5 in radians or cycles from the uncombined carriers. Both entries must use the same unit. Do not substitute metres into the ratio. The ionosphere-free combination destroys the diagnostic ratio. Passing the delay class rejects only ionospheric and digital-artifact explanations. Troposphere, multipath, and clock error are also delay-class. A candidate that matches no class is not promoted on chromaticity. Switch 2 cannot fire on a null.

Step 12. Adjudicate the switches

The book does not mark a switch passed because a field was left at its default. Record the call you are making: null holds, candidate passed, fired, or not applicable on a null. Candidate passed requires the uncombined chromatic entry. Details and blast radii remain on the Switches screen.

Step 13. Close and publish

- Open Book and use the campaign-close export prompt to download JSON.
- Export Markdown for a human-readable campaign note and move both files off the handset.
- Raw IQ remains on the nodes.
- Report three distinct values for each chain: predicted budget | measured residual floor | reported bound.
- Chain S uses the measured double-difference floor. Chain C uses the per-baseline maximum from Step 5.
- Publish a null campaign. The absence of an excess does not make the record disposable.

A.4 Bench, Book, and record care

Bench is a calculator. It never appends the campaign log. Tools: phase stability $\sigma_\varphi = 2\pi f \tau \sigma_y$; light time; stationary floor (Beta null and the $L = 86,400$ trap); weak common phase $\varphi \approx \sigma \sqrt{r}$ beside exact inversion; chromatic class; ionospheric phase (0.1 TECU example, not a forecast); per-dump phase; guard band; sphere baseline (mean-Earth haversine, not a geodetic reduction); campaign RSS; Rev 3.6 identity self-check (a failed row means the app is wrong).

Book archive. Export the book downloads `dslv-zpdi-vacuum-book.json` with every local campaign, tagged for this revision. Export this campaign downloads one campaign as JSON. Markdown downloads a human-readable note. Import JSON appends records; it does not overwrite a campaign already on the phone. Invalid files are refused. Field notes append; older entries are not edited. Remove this campaign deletes one local record after a second confirmation. Export first. Deleting browser site data can destroy the only copy held by this app.

Mandatory JSON prompts: registry freeze (after the external anchor ID is recorded); amendment re-freeze; calibration append; campaign close. Move the archive off the phone by USB or a storage share you trust. Treat JSON as the registry: it may contain operator names, site coordinates, freeze hashes, and external anchor IDs.

A.5 Kill switches (short form)

1. Array residual. Fires when Analysis A agrees with its null and Analysis B's bootstrap maximum agrees with the time-slide null, evaluated per chain. Blast radius: that chain's hypothesis. The array remains a disciplined SDR network.

2. Chromaticity. Fires on the uncombined L1/L5 pair failing the registered delay class. Passing is necessary, not sufficient. Blast radius: the candidate. Cannot fire on a null.

3. Pipeline non-recovery. Injection miss, failed return to floor, FAR disagreement, or broken common-clock calibration. Blast radius: the campaign. No science claim.

6. Simultaneity convention. Excess unstable under independent time transfer or calibration substitution. Blast radius: any non-local reading of that dataset. Load-bearing on Chain C.

A.6 Troubleshooting

Symptom	Check
Blank screen after install	Allow JIT for this origin only; confirm network permission; reopen the published host, not a preview origin.
Install does nothing	Drag the shortcut from the placement sheet onto the home screen. Do not use Incognito.
Missing campaign after reopen	Wrong origin or site data cleared. There is no cloud copy.
Next disabled	The current Walk gate is unsatisfied. Back is always allowed.
Formula identity fail	App bug. Not physics. Do not log it as a residual.
Cannot freeze	Registry incomplete, calibration/bias hash missing, or fewer than the required identified nodes.
Cannot claim candidate	Need freeze + external anchor + chromatic entry + injection line.

End of submission compilation. Joseph Robert Fross · Resonant Genesis LLC · DSLV-ZPDI Rev 3.6 · 28 September 2026. This packet contains the mission cover, the instrument white paper, and the field-book install/user guide. Interpretive supplements S1/S2 are withheld from review.